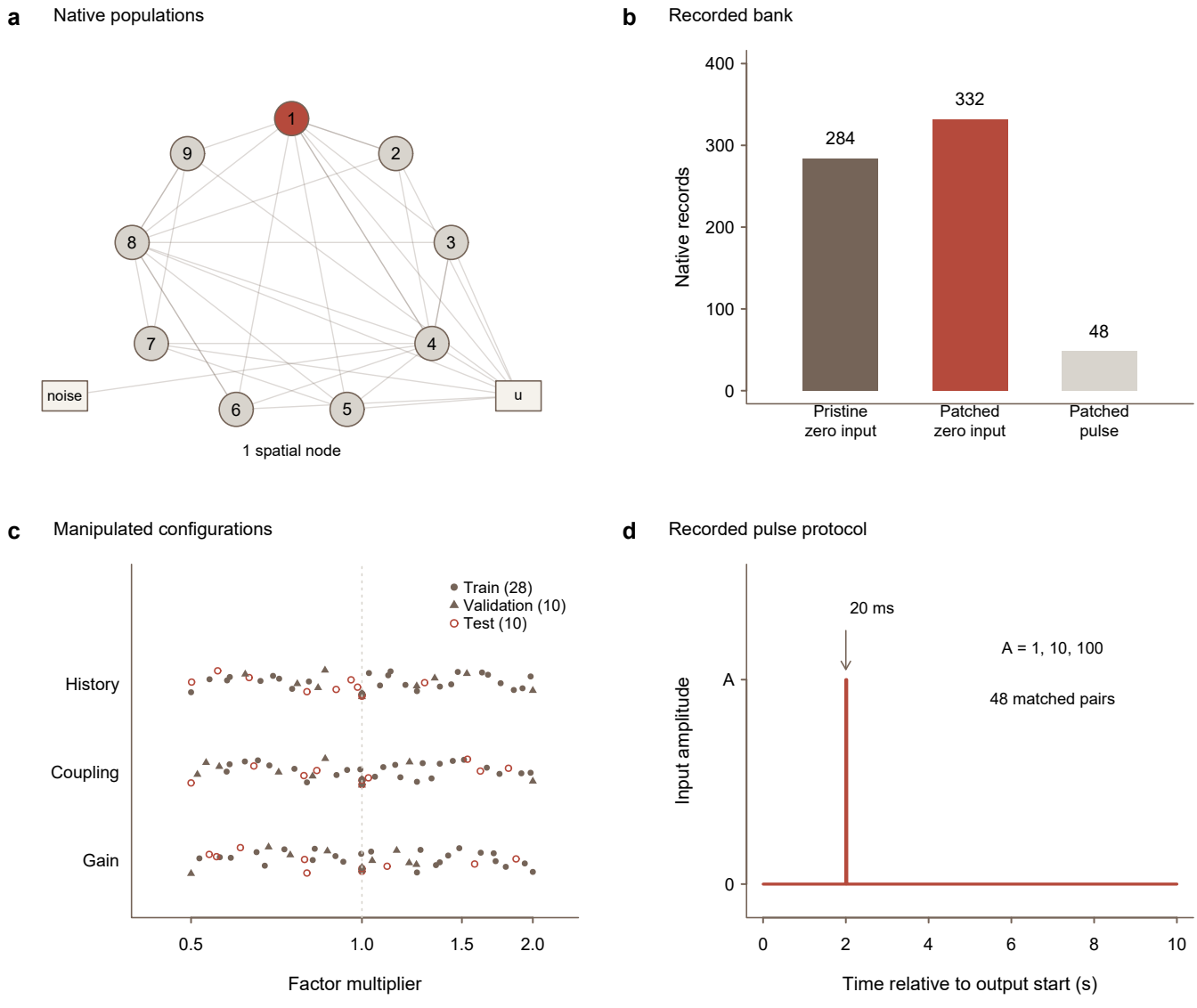
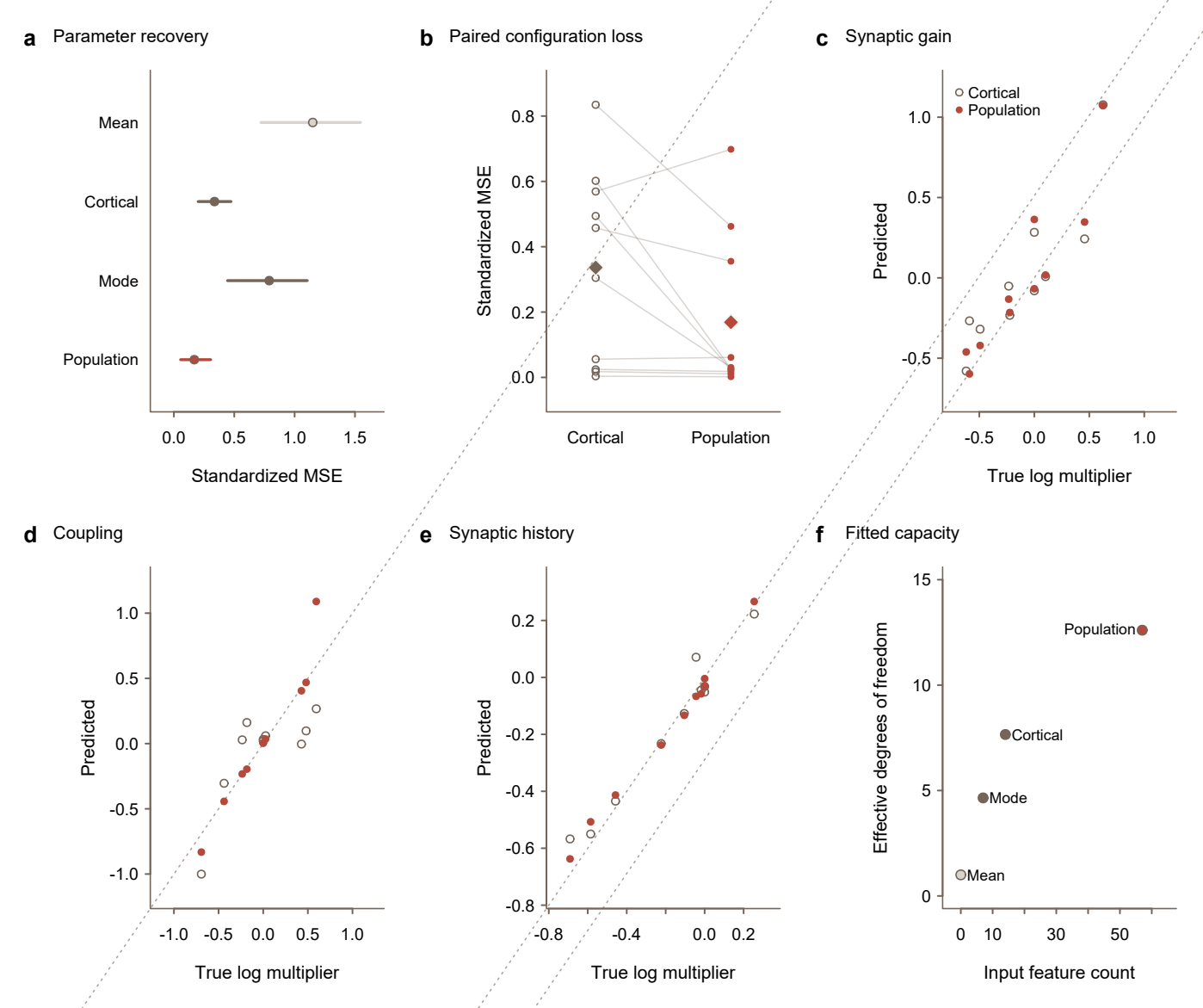
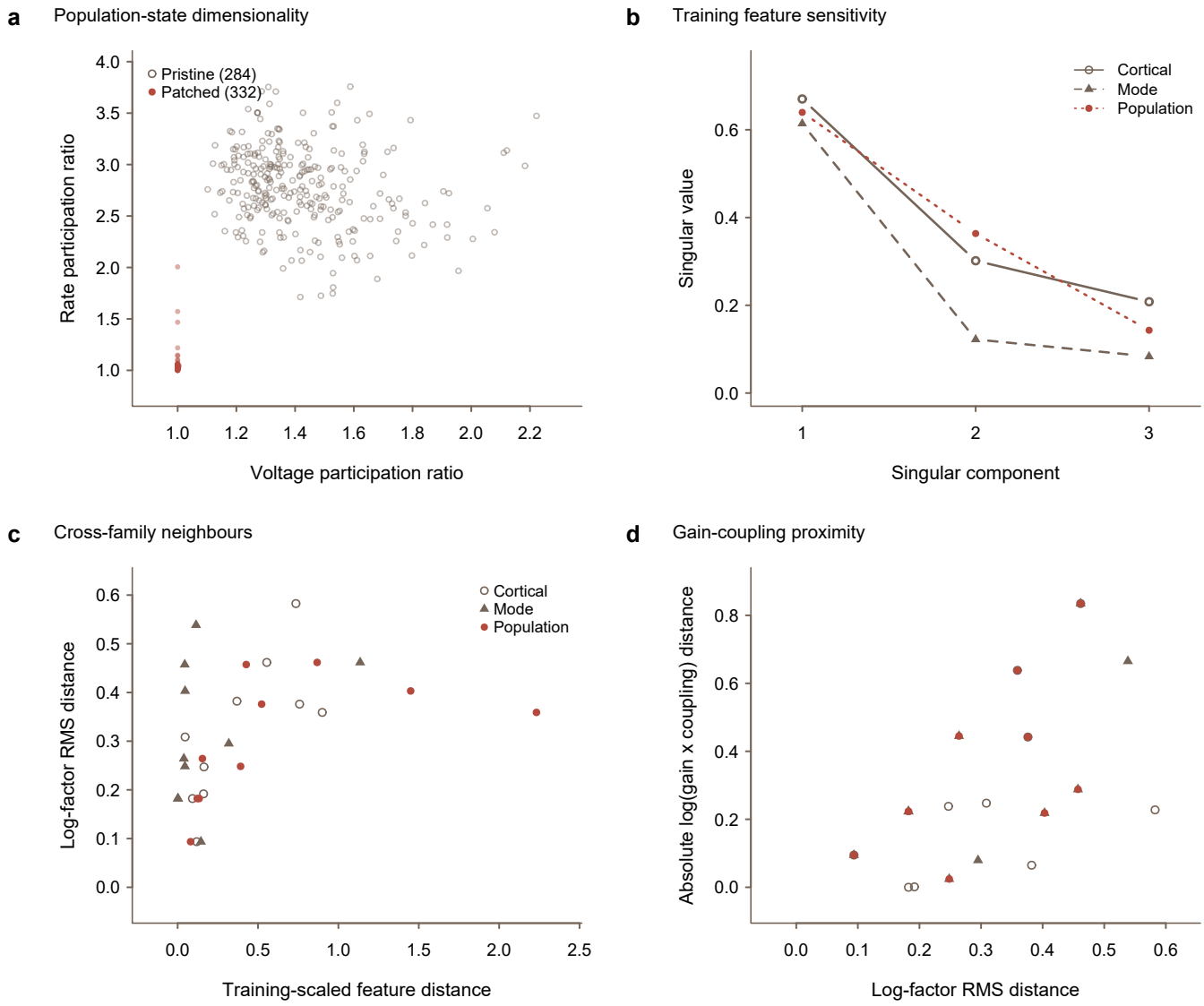
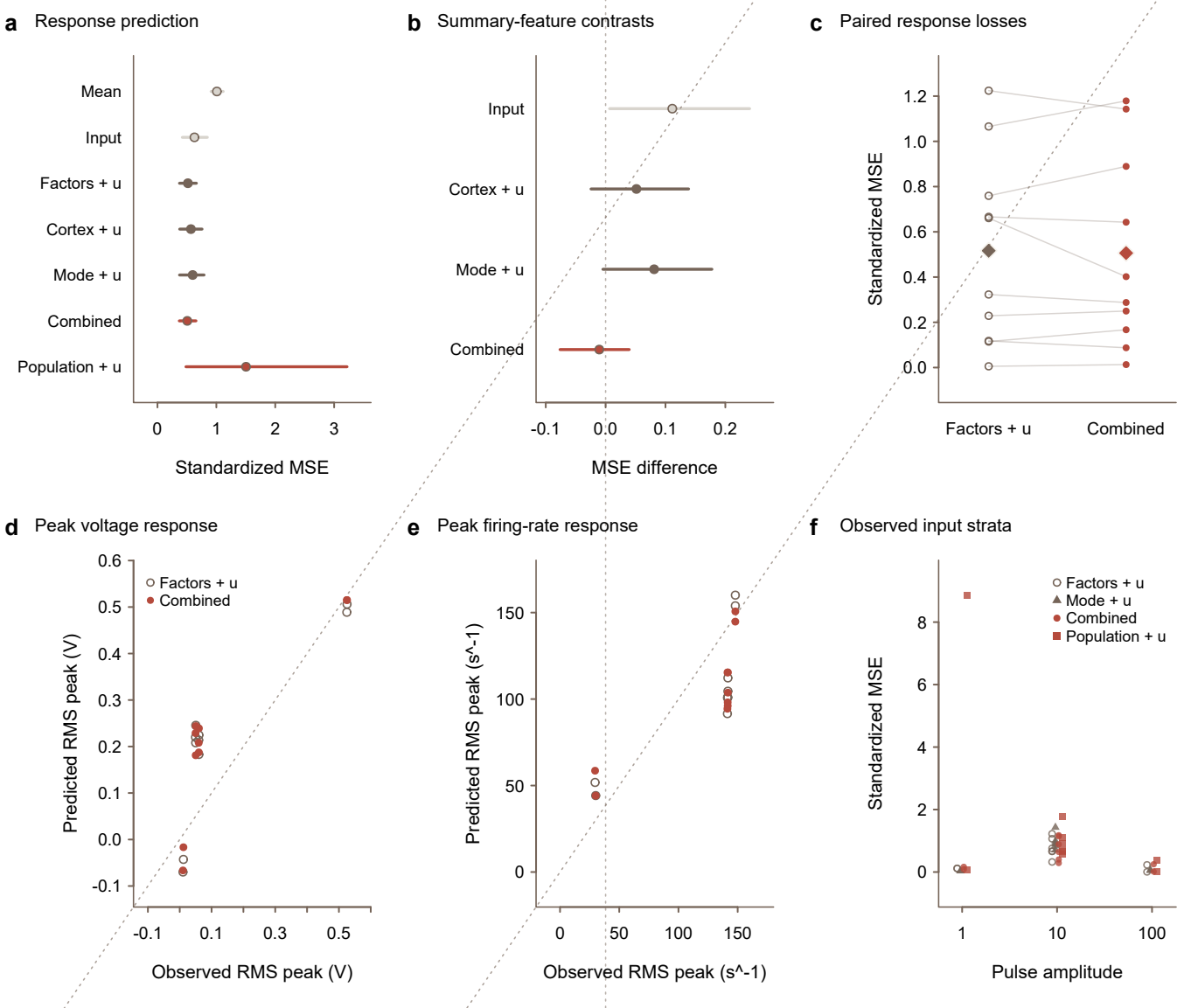
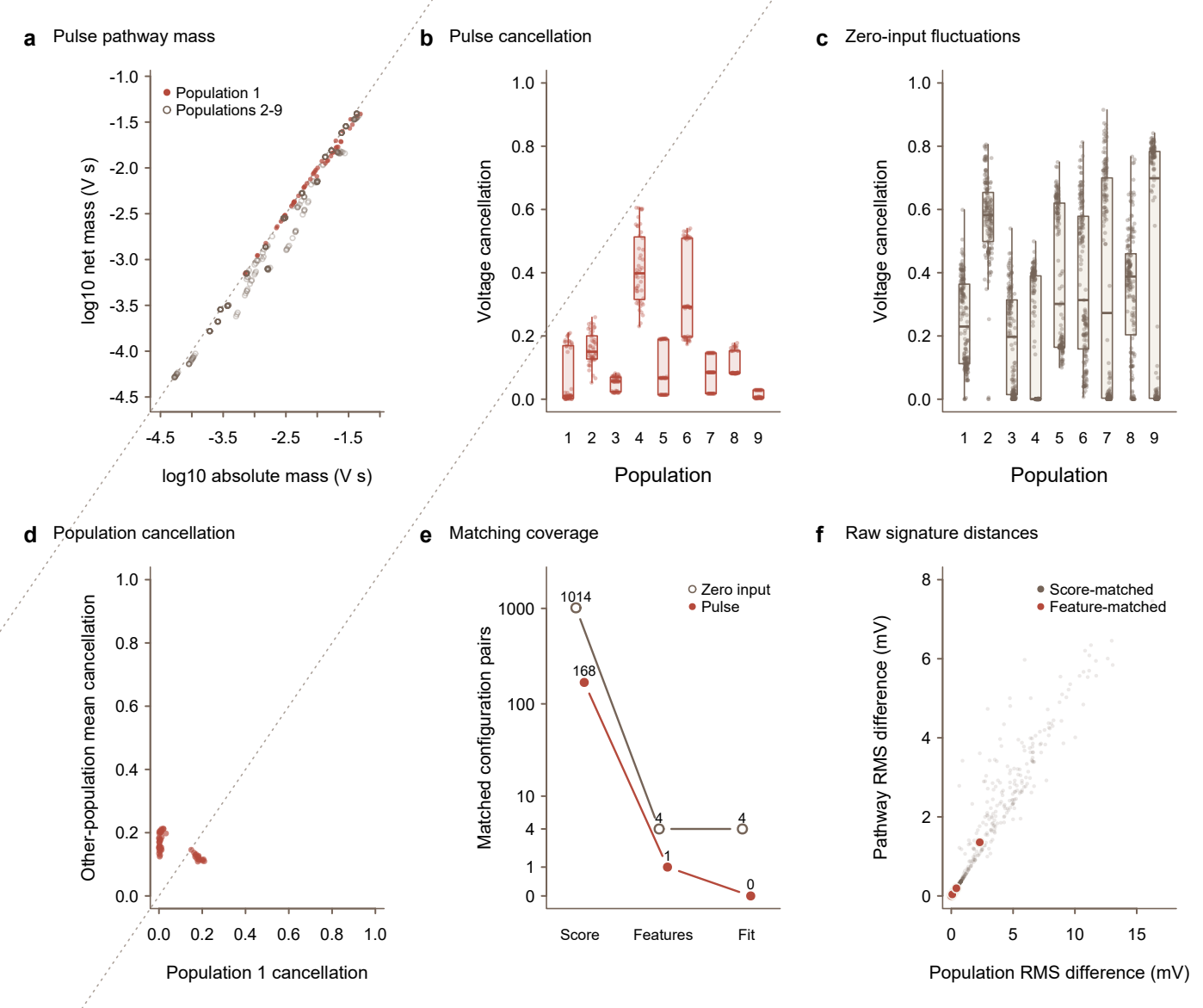

a
Native populations
1
2
3
4
5
6
7
8
9
noise
u
1 spatial node
b
Recorded bank
Native records
400
300
200
100
0
284
332
48
Pristine
zero input
Patched
zero input
Patched
pulse
c
Manipulated configurations
Train (28)
Validation (10)
Test (10)
History
Coupling
Gain
0.5
1.0
1.5
2.0
Factor multiplier
d
Recorded pulse protocol
20 ms
A = 1, 10, 100
48 matched pairs
Input amplitude
A
0
0
2
4
6
8
10
Time relative to output start (s)


a
Parameter recovery
Mean
Cortical
Mode
Population
0.0
0.5
1.0
1.5
Standardized MSE
b
Paired configuration loss
Standardized MSE
0.8
0.6
0.4
0.2
0.0
Cortical
Population
c
Synaptic gain
Cortical
Population
Predicted
1.0
0.5
0.0
-0.5
-0.5
0.0
0.5
1.0
True log multiplier
d
Coupling
Predicted
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
True log multiplier
e
Synaptic history
Predicted
0.2
0.0
-0.2
-0.4
-0.6
-0.8
-0.8
-0.4
0.0
0.2
True log multiplier
f
Fitted capacity
Effective degrees of freedom
15
10
5
0
Population
Cortical
Mode
Mean
0
10
30
50
Input feature count


a
Population-state dimensionality
Pristine (284)
Patched (332)
Rate participation ratio
Voltage participation ratio
b
Training feature sensitivity
Cortical
Mode
Population
Singular value
Singular component
c
Cross-family neighbours
Cortical
Mode
Population
Log-factor RMS distance
Training-scaled feature distance
d
Gain-coupling proximity
Absolute log(gain x coupling) distance
Log-factor RMS distance


a
Response prediction
Mean
Input
Factors + u
Cortex + u
Mode + u
Combined
Population + u
0
1
2
3
Standardized MSE
b
Summary-feature contrasts
Input
Cortex + u
Mode + u
Combined
-0.1
0.0
0.1
0.2
MSE difference
c
Paired response losses
Standardized MSE
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Factors + u
Combined
d
Peak voltage response
Factors + u
Combined
Predicted RMS peak (V)
-0.1
0.1
0.3
0.5
Observed RMS peak (V)
e
Peak firing-rate response
Predicted RMS peak (s^-1)
0
50
100
150
Observed RMS peak (s^-1)
f
Observed input strata
Factors + u
Mode + u
Combined
Population + u
Standardized MSE
0
2
4
6
8
1
10
100
Pulse amplitude


a
Pulse pathway mass
Population 1
Populations 2-9
log10 net mass (V s)
log10 absolute mass (V s)
b
Pulse cancellation
Voltage cancellation
Population
c
Zero-input fluctuations
Voltage cancellation
Population
d
Population cancellation
Other-population mean cancellation
Population 1 cancellation
e
Matching coverage
Zero input
Pulse
Matched configuration pairs
1014
168
4
4
1
0
Score
Features
Fit
f
Raw signature distances
Score-matched
Feature-matched
Pathway RMS difference (mV)
Population RMS difference (mV)